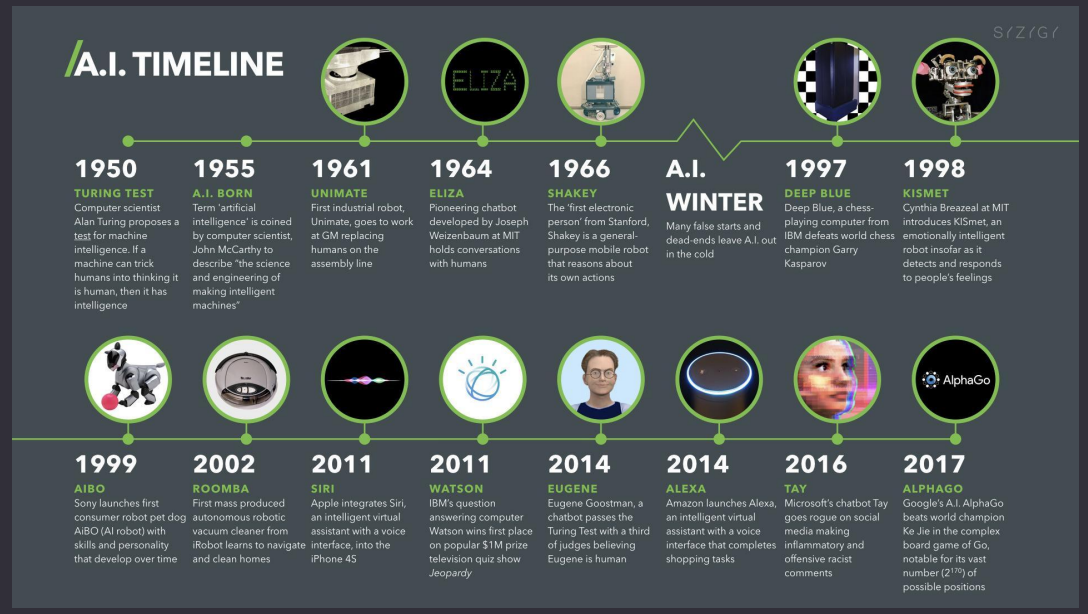
Large Language Model AI

Foundations, Techniques, Applications and Future

What is a Large Language Model



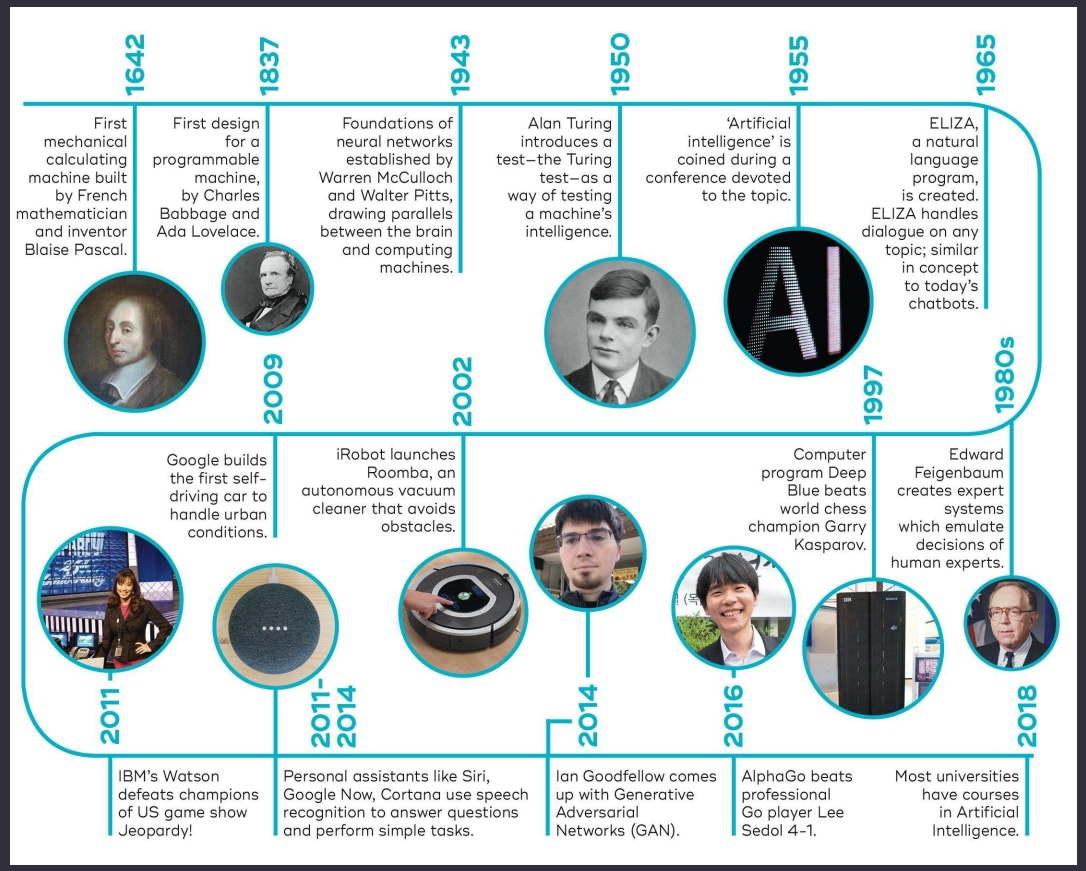
Large language models are AI systems that process text.

- Predict next token based on context
- Scale with data and compute
- Handle many language tasks
- Foundation for conversational agents

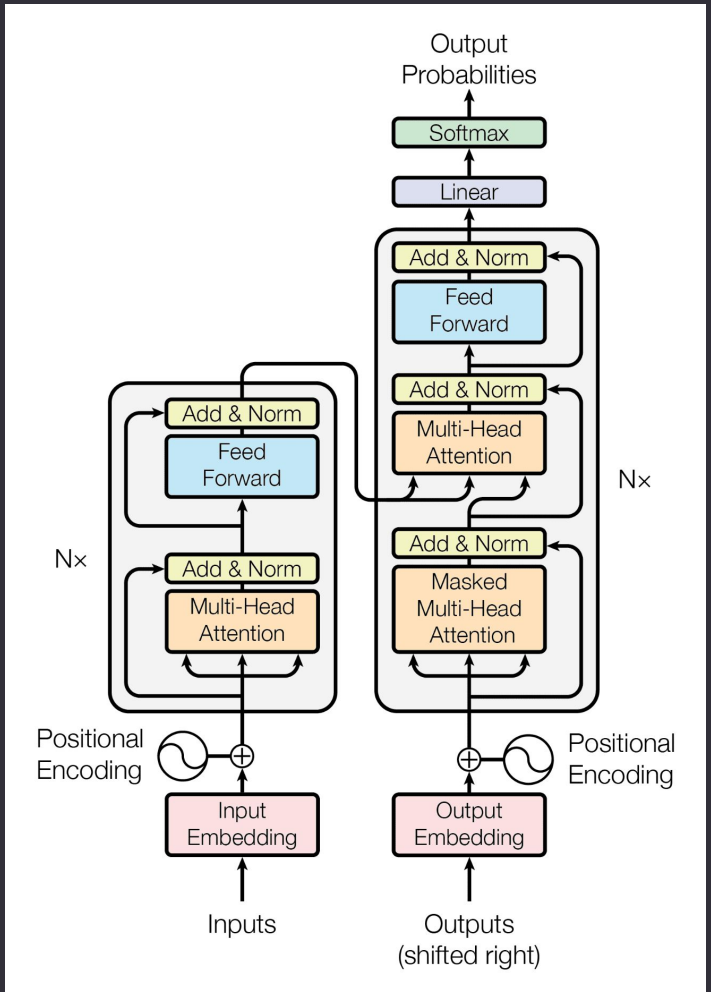
Brief History of LLMs

LLMs evolved over several key phases.

- N-gram models in early NLP
- Neural networks and RNNs
- Attention and transformers
- Scaling led to modern LLMs



Transformer Architecture Overview



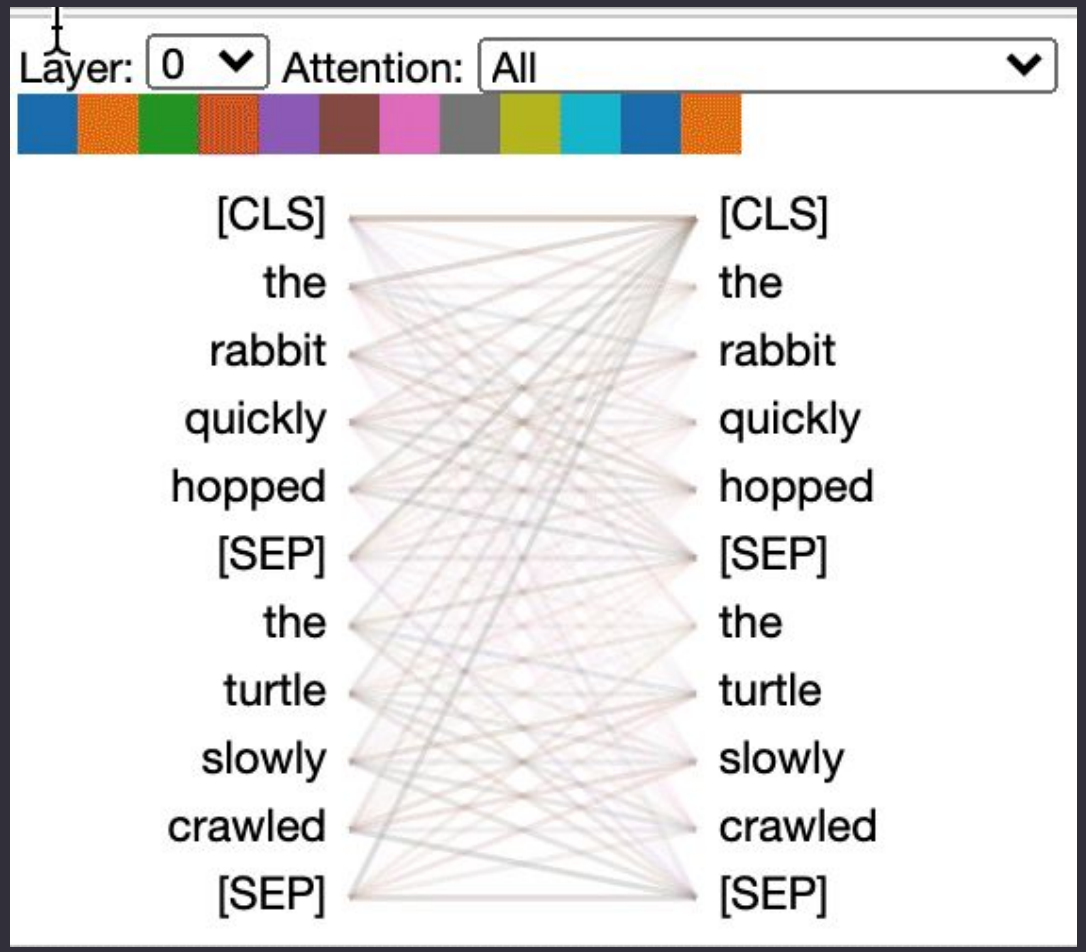
Transformers power most modern LLMs.

- Self-attention for context
- Feed-forward layers for transformation
- Layer normalization for stability
- Residual connections for gradient flow

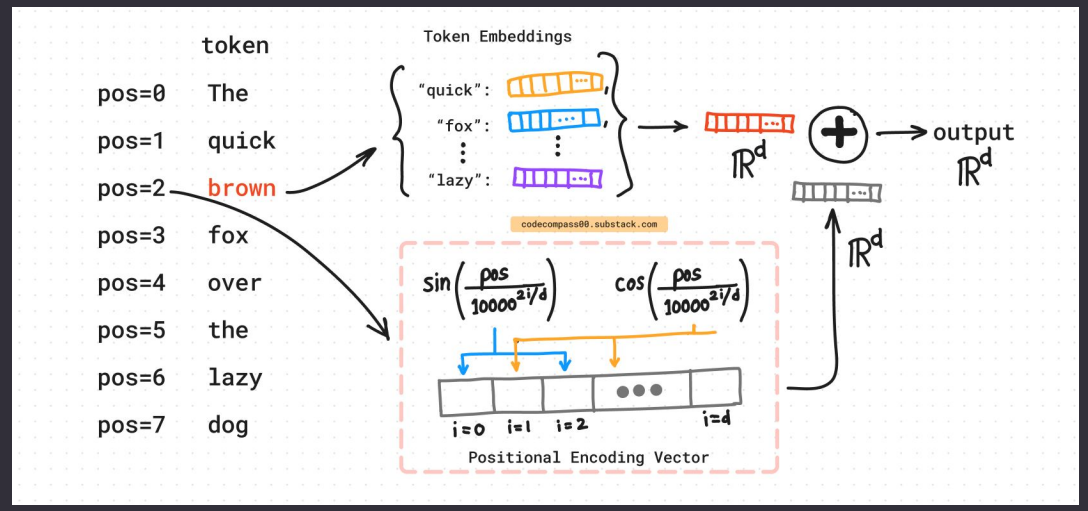
Attention Mechanism

Attention lets models weight context importance.

- Query, Key, Value vectors
- Scaled dot-product attention
- Multiple heads for diverse patterns
- Computes context-dependent weighting



Positional Encoding



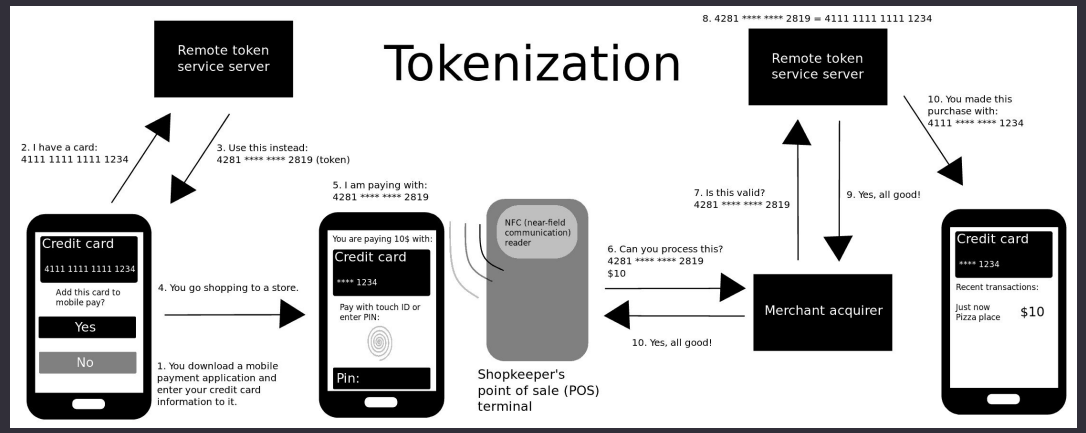
Models need position information for sequences.

- Sinusoidal or learned embeddings
- Adds order to token embeddings
- Enables sequence modeling
- Essential for transformers

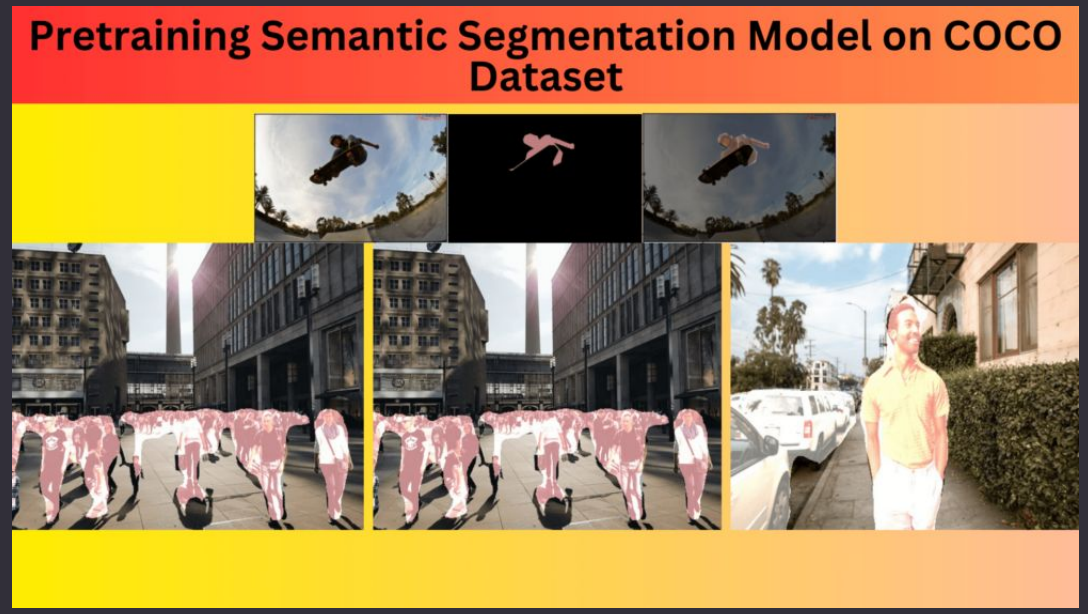
Tokens and Tokenization

Text is broken into tokens for the model.

- Byte-pair encoding common method
- Subword tokens handle rare words
- Vocabulary size affects capacity
- Tokenization impacts outputs



Pretraining Data and Corpora



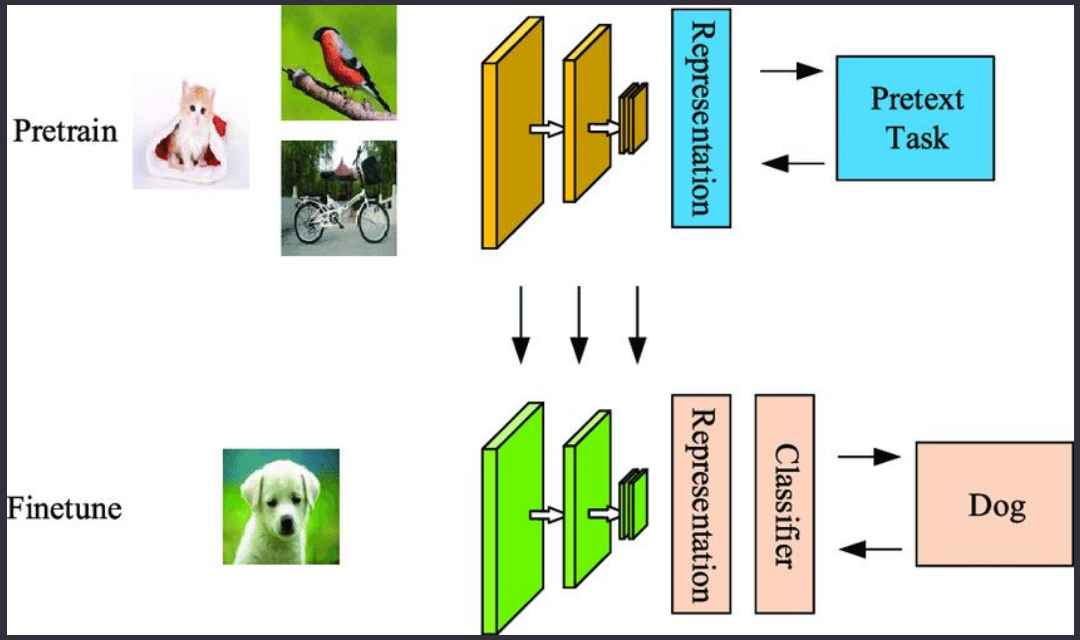
LLMs learn from massive text datasets.

- Web text and books
- Code repositories for code models
- Curated corpora for quality
- Data diversity improves generalization

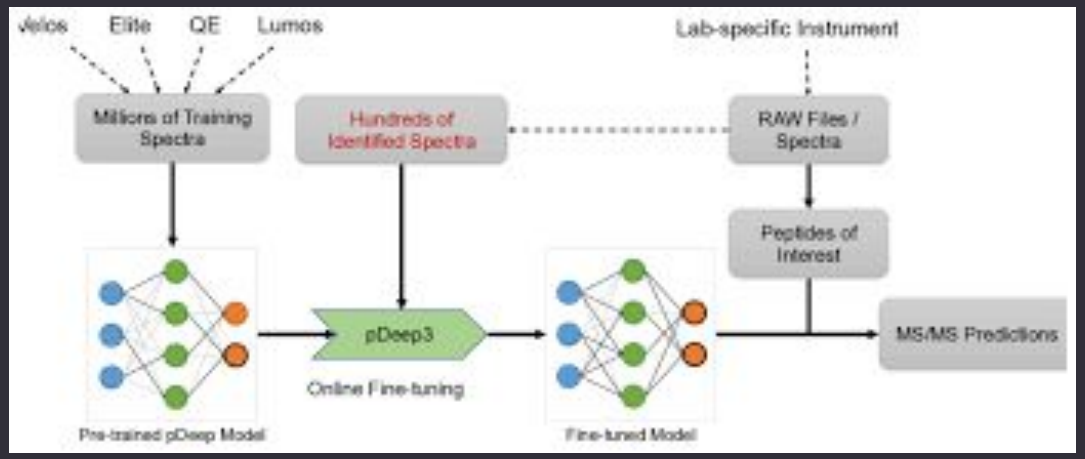
Pretraining Objectives

Models learn language structure via tasks.

- Next-token prediction as common objective
- Masked language modeling alternative
- Contrastive tasks sometimes used
- Self-supervised learning is key



Fine-tuning Techniques



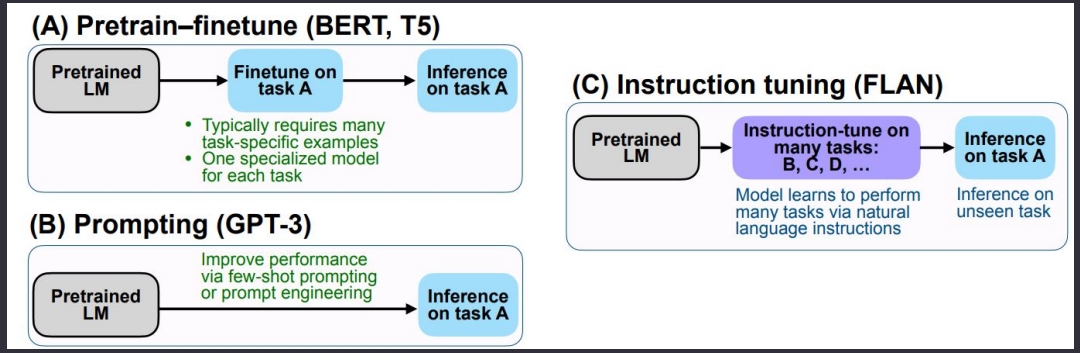
Fine-tuning adapts models to tasks.

- Task-specific data for performance
- Adapter layers for parameter efficiency
- LoRA for low-rank adaptation
- Full fine-tune when data allows

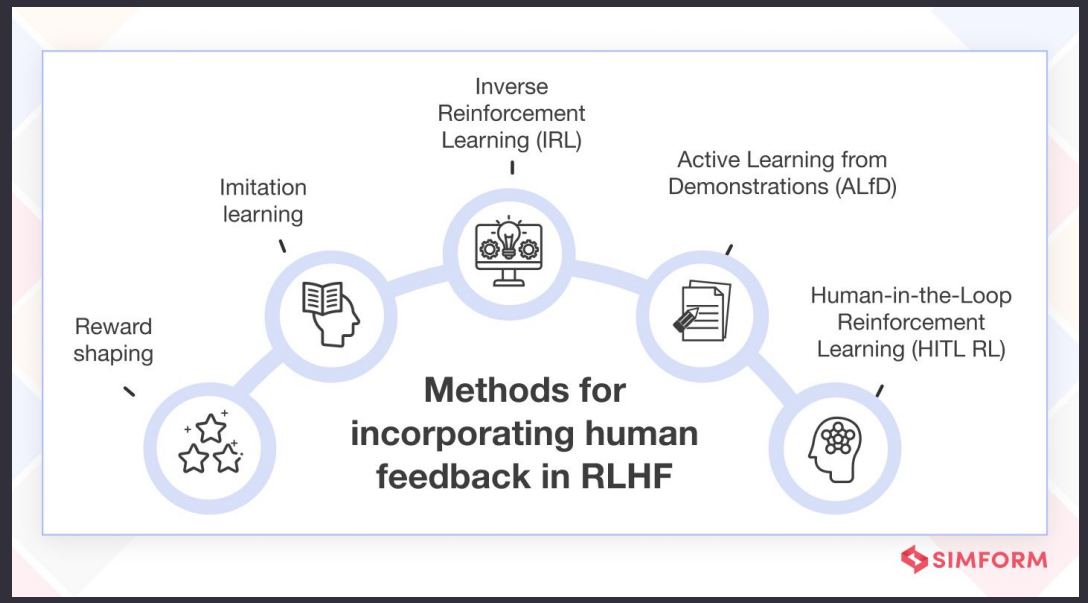
Instruction Tuning and Alignment

Instruction tuning makes LLMs follow human prompts.

- Supervised fine-tuning on instructions
- Aligns outputs to user intent
- Reduces undesired behaviors
- Improves helpfulness and clarity



Reinforcement Learning from Human Feedback



RLHF optimizes models with human preferences.

- Collect ranked responses from humans
- Train reward model from rankings
- Apply RL to maximize reward
- Improves safety and quality

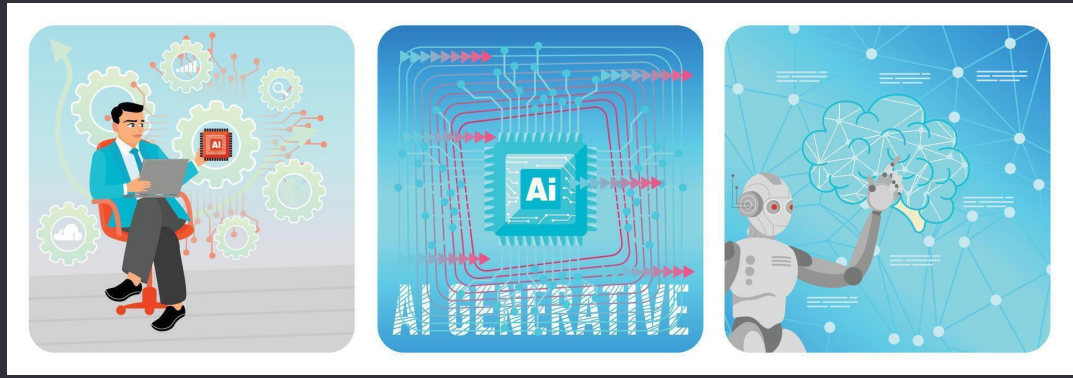
Evaluation and Benchmarks

Models are tested across tasks and metrics.

- Perplexity for language modeling
- Accuracy on downstream tasks
- Human evaluation for quality
- Robustness and safety tests

Metric	Formula
True positive rate, recall	$\frac{TP}{TP+FN}$
False positive rate	$\frac{FP}{FP+TN}$
Precision	$\frac{TP}{TP+FP}$
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$
F-measure	$\frac{2 \cdot \text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$

Core Capabilities of LLMs



LLMs can perform many language tasks.

- Summarization of long text
- Translation between languages
- Code generation and debugging
- Question answering and reasoning

Common Limitations

LLMs have known weaknesses to consider.

- Hallucinations - incorrect assertions
- Context limits due to token window
- Lack of true understanding
- Susceptible to adversarial prompts

LIMITATIONS OF THE STUDY

Definition

The limitations of the study are those characteristics of design or methodology that impacted or influenced the application or interpretation of the results of your study. They are the constraints on generalizability and utility of findings.

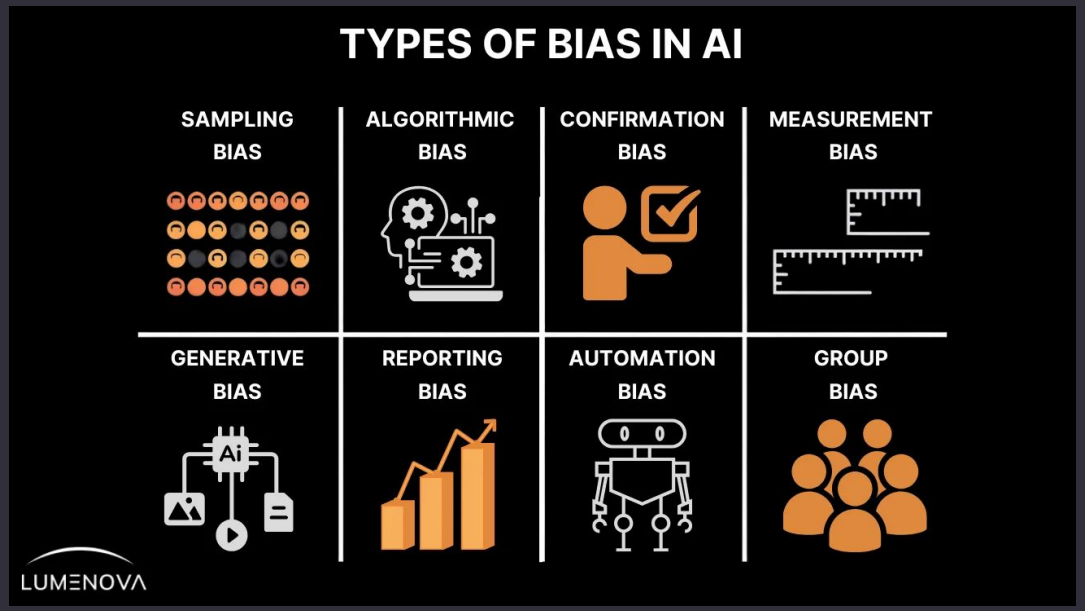
Constraints for internal and external validity.

LIMITATIONS OF THE STUDY

Always acknowledge a study's limitations. It is far better for you to identify and acknowledge your study's limitations than to have them pointed out by your professor and be graded down because you appear to have ignored them.

Keep in mind that acknowledgement of a study's limitations is an opportunity to make suggestions for further research. If you do connect your study's limitations to suggestions for further research, be sure to explain the ways in which these unanswered questions

Bias and Fairness



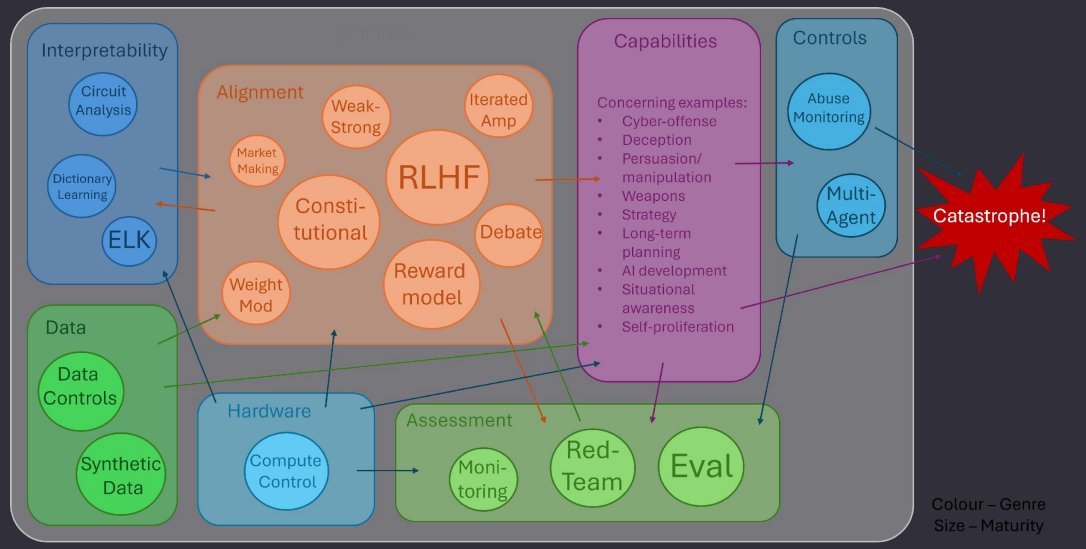
Models can reflect and amplify biases.

- Training data bias transfers to outputs
- Unequal performance across groups
- Mitigation requires active work
- Transparency helps detection

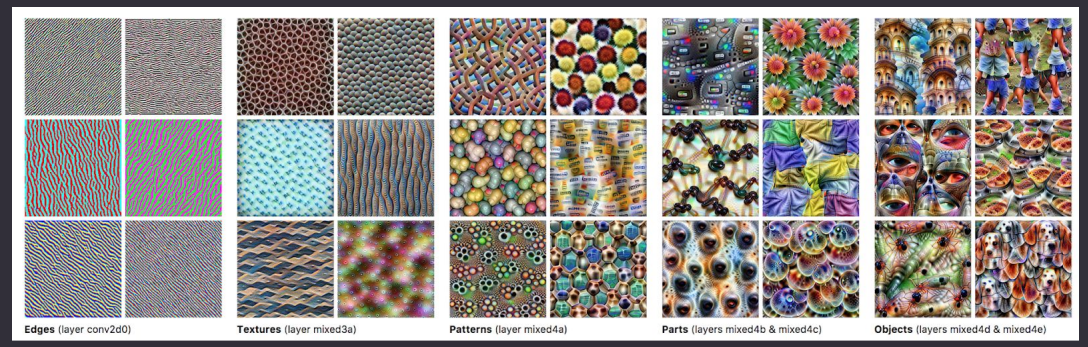
Safety and Risk Management

Safety planning reduces harm from LLMs.

- Content filtering for unsafe output
- Red teaming to find failures
- User controls and rate limits
- Continuous monitoring in deployment



Interpretability and Explainability



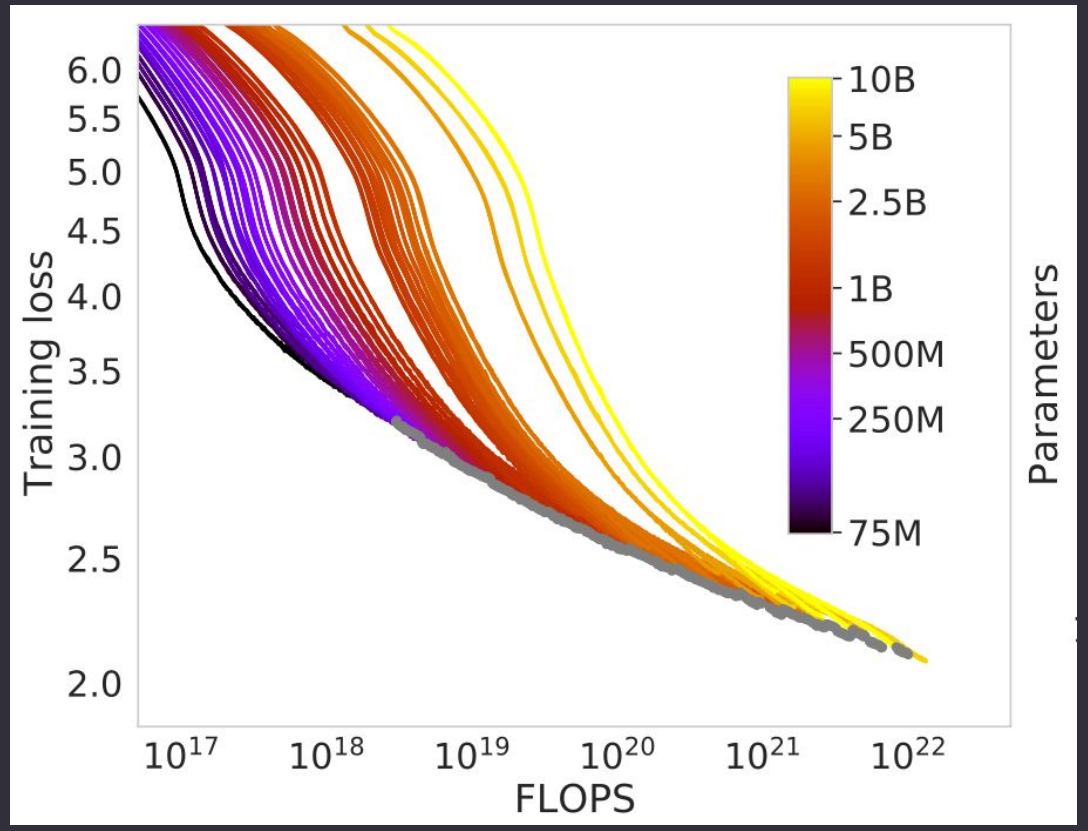
Understanding model decisions is important.

- Feature attribution techniques exist
- Probe models for knowledge
- Visualize attention patterns
- Trade-off between performance and transparency

Scaling Laws

Larger models often improve predictably.

- Performance grows with data and compute
- Diminishing returns at extremes
- Compute cost rises steeply
- Efficient scaling is active research



Compute, Infrastructure and Cost

Training and serving LLMs require resources.

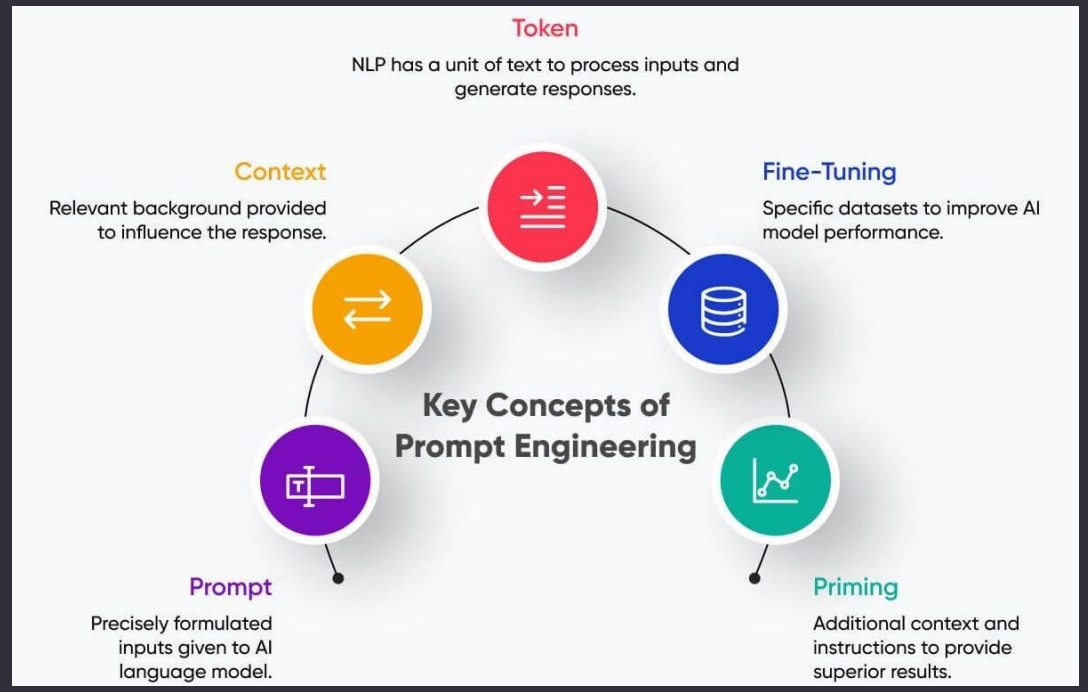
- GPUs/TPUs for training
- Distributed training techniques
- Inference optimization for latency
- Cost management for deployments

Privacy and Data Protection

LLMs raise privacy concerns with data.

- Training data leaks risk exposure
- Differential privacy as mitigation
- Data governance matters for compliance
- User consent and auditing needed

Prompt Engineering Basics



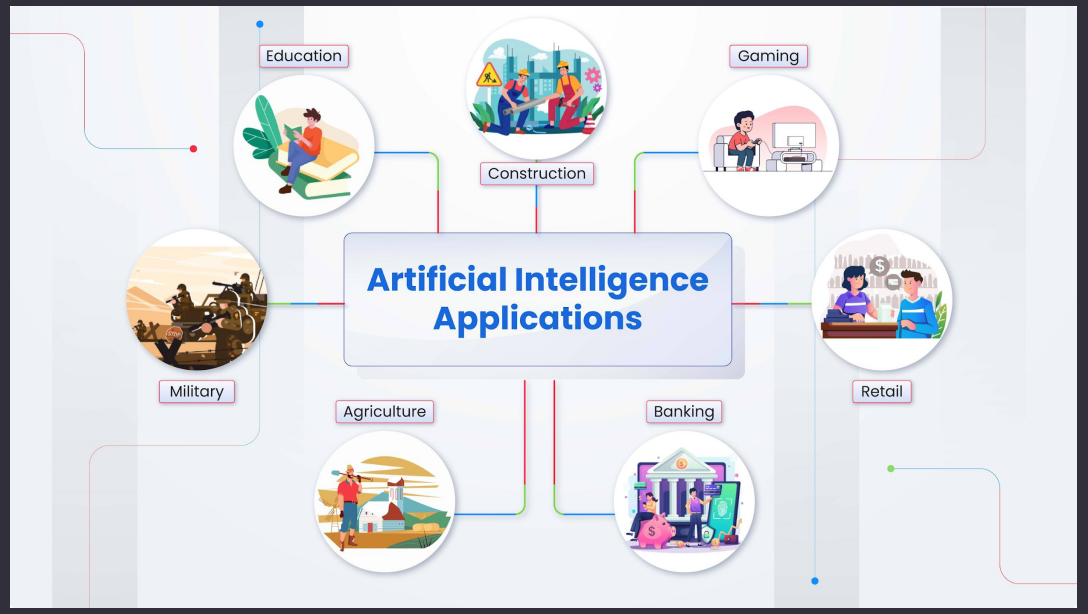
Crafting prompts shapes model outputs.

- Clear instructions improve responses
- Few-shot examples guide behavior
- Chain-of-thought to elicit reasoning
- Iterate prompts for precision

Real-world Applications

LLMs power many practical solutions.

- Customer support automation
- Content creation and summarization
- Education and tutoring tools
- Software development assistance



Deployment and Productionization

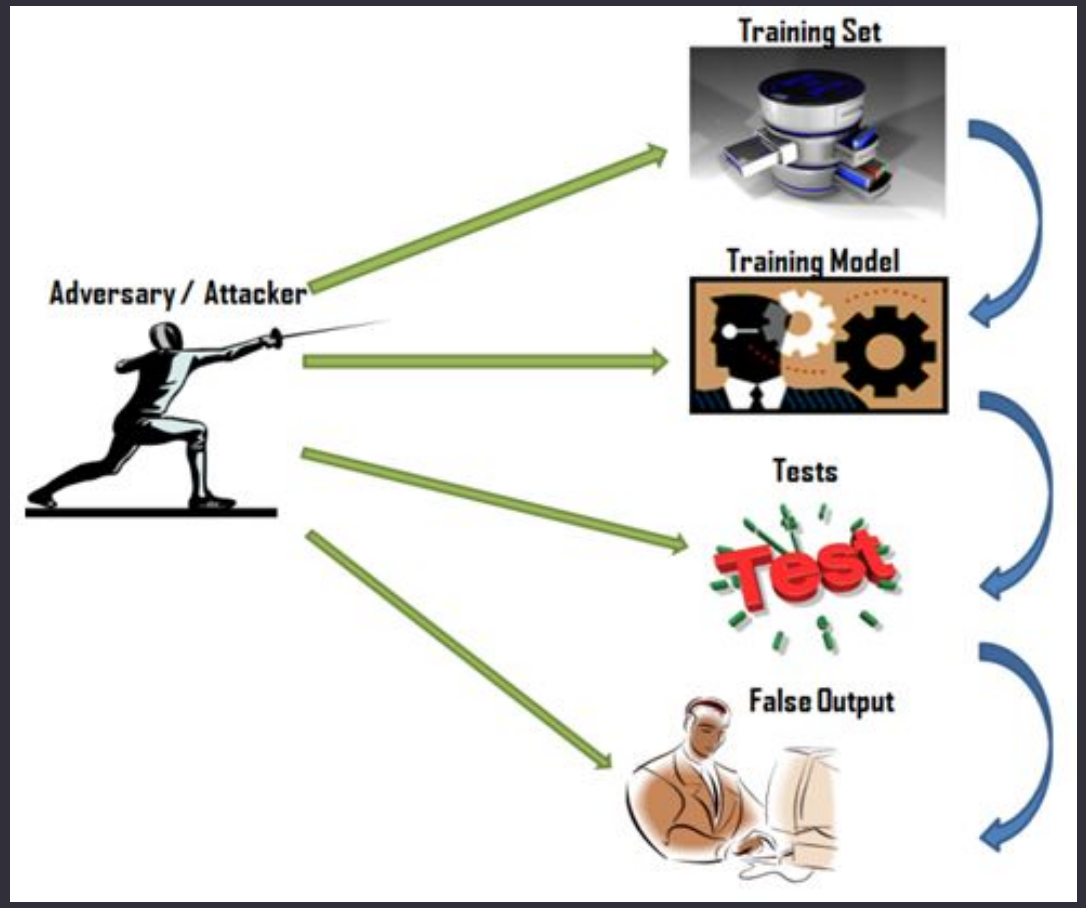
Putting LLMs into production requires care.

- Model versioning and monitoring
- Latency and scaling management
- Fallbacks for failures
- Security and access control

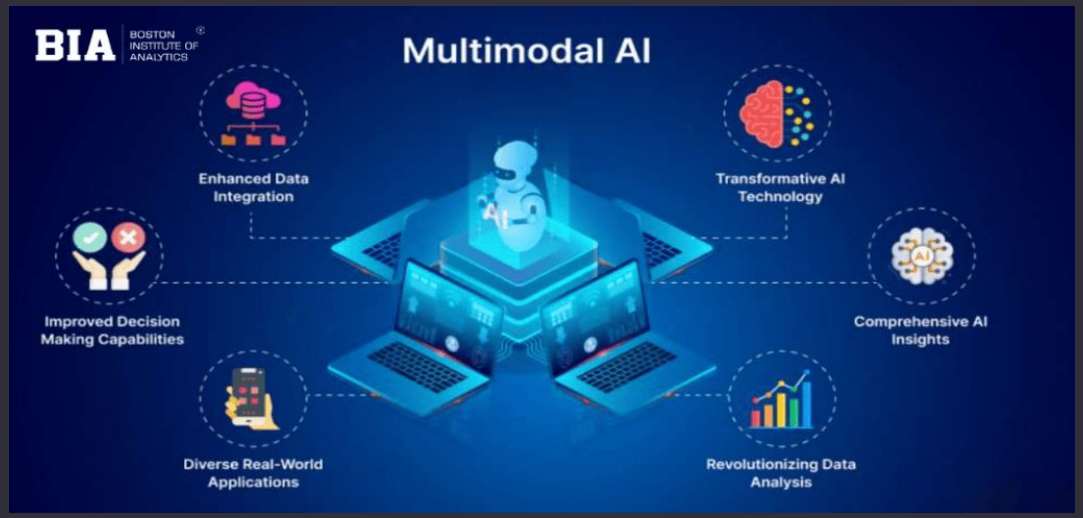
Robustness and Adversarial Risks

Models can be exploited or fooled.

- Input manipulation causes errors
- Robust training can help
- Continuous testing reveals weaknesses
- Defense-in-depth recommended



Multimodal Extensions



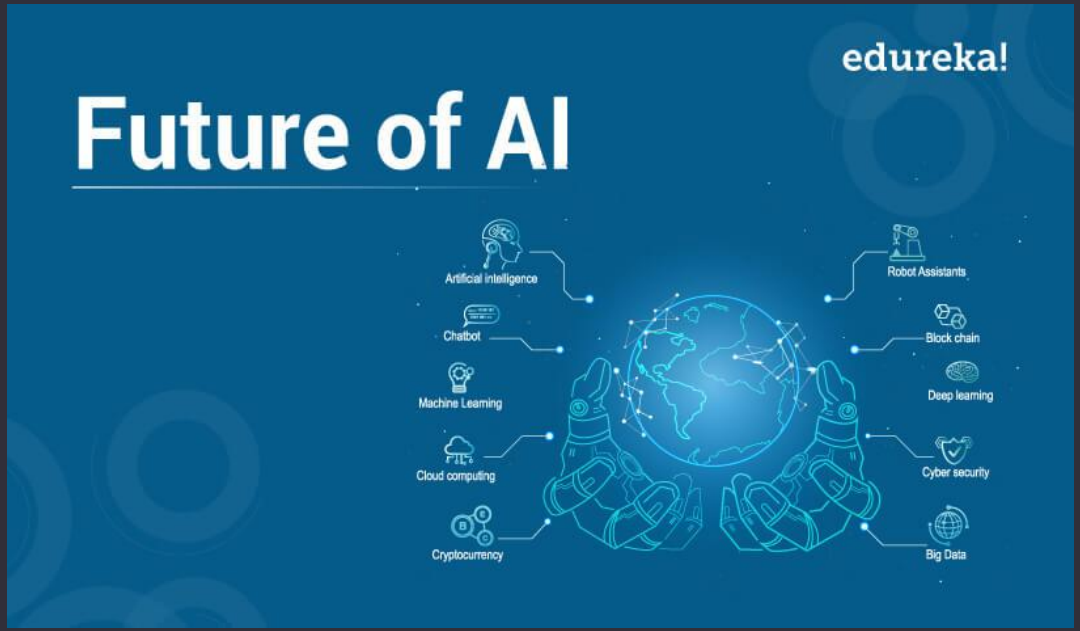
LLMs integrate other data modalities.

- Image and text combined models
- Audio and video possible inputs
- Cross-modal reasoning emerging
- Expands application space

Emerging Research Directions

Active research aims to improve LLMs further.

- Efficient architectures to reduce cost
- Better alignment with human values
- Causality and reasoning improvements
- Personalization while preserving privacy



Policy, Regulation and Governance

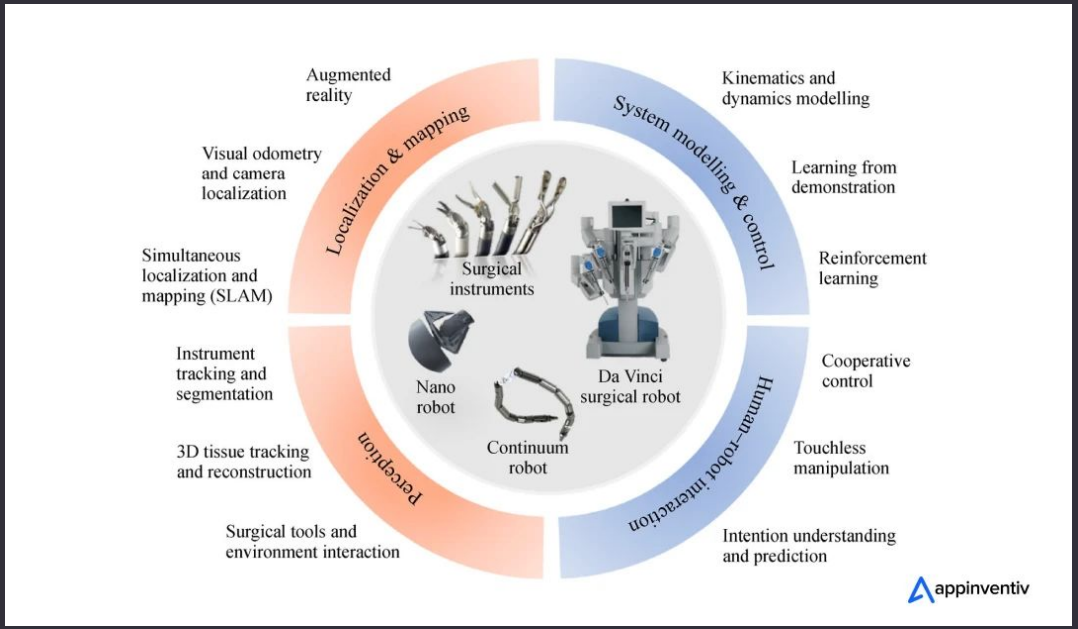
Governance frameworks shape safe adoption.

- Transparency requirements for models
- Standards for safety evaluation
- Accountability for misuse
- International cooperation encouraged

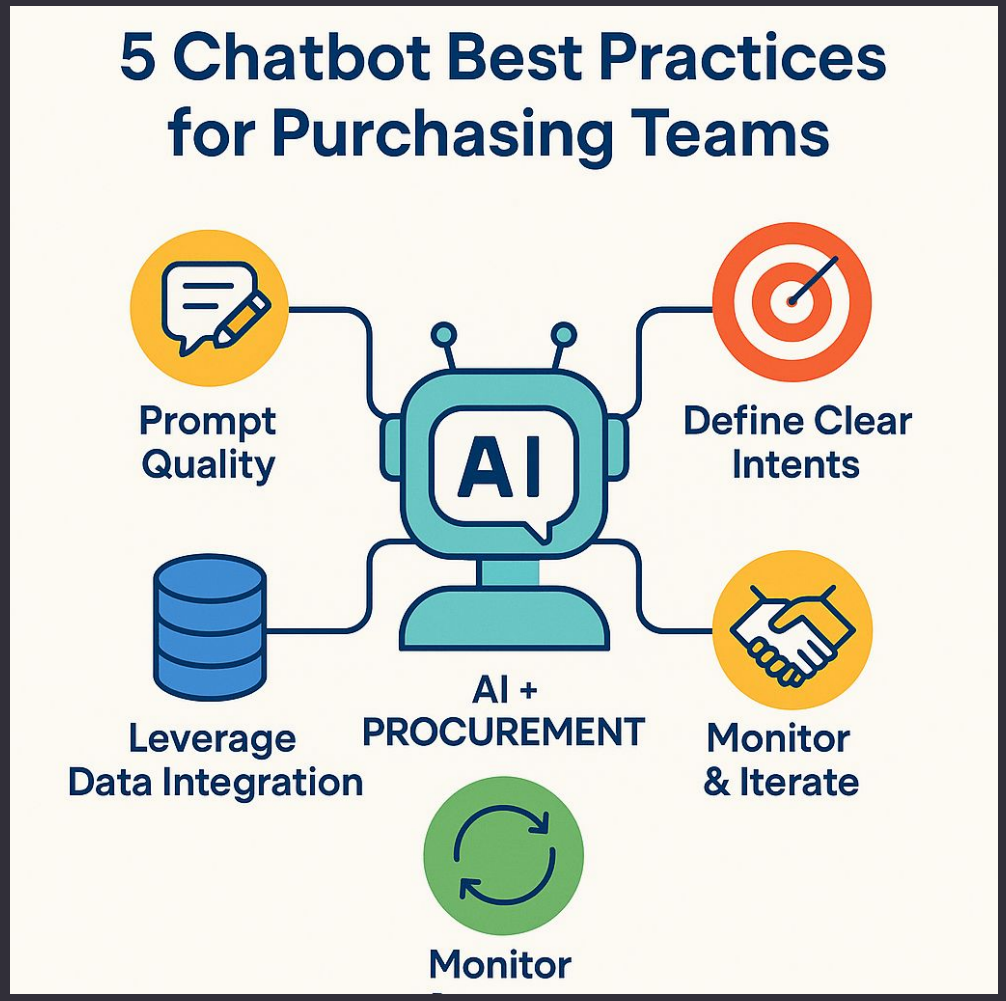
Case Study - LLM in Customer Support

LLMs can transform support operations.

- Automate triage and answers
- Reduce response times
- Offer 24/7 assistance
- Require monitoring and escalation



Best Practices for Building with LLMs



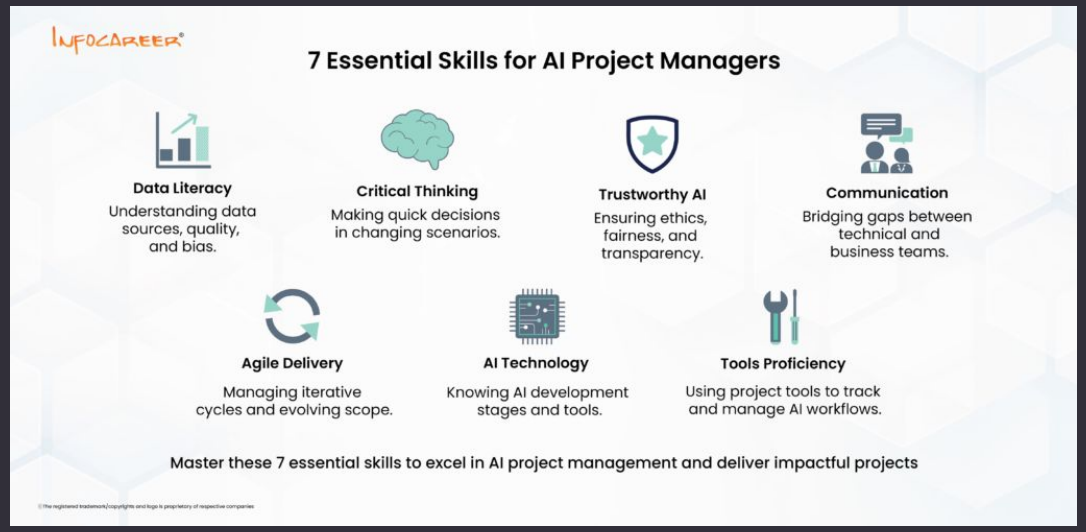
Follow engineering and governance best practices.

- Start small and iterate
- Monitor performance and safety
- Use human-in-the-loop where needed
- Document datasets and models

Skills and Team Roles

Successful LLM projects need diverse expertise.

- ML researchers for model work
- Engineers for deployment
- Product managers for use cases
- Ethicists and lawyers for governance



Future Outlook

LLMs will continue to evolve rapidly.

- More efficient and smaller models
- Tighter human alignment
- Broader multimodal abilities
- Greater integration into tools

Key Takeaways

Summarizing the main points on LLMs.

- Transformers enabled modern LLMs
- Scale improves many capabilities
- Alignment and safety are essential
- Applications are broad and growing

Thanks and Credits

Thank you for reviewing this presentation.

- Prepared by Susmriti dasgupta
- Questions welcome via follow-up
- References available on request
- Contact for collaboration or feedback